

Amro Abdrabo

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EDUCATION

ETH Zürich <i>MSc Computer Science</i> MSc coursework: Info Security Lab, System Security, Digital Signatures, Program Analysis for Security, Computational Statistics, German C1.	Sept 2020 — Nov 2024 <i>Zurich, Switzerland</i>
École Polytechnique Fédérale de Lausanne <i>CMS (5.46) & BSc Computer Science (5.20)</i>	Sept 2016 — Sept 2020 <i>Lausanne, Switzerland</i>
Dawson College <i>Cégep (96% Average Grade)</i>	Aug 2014 — June 2016 <i>Montréal, Canada</i>

WORK EXPERIENCE

Graduate Assistant HEC - UNIL - Assistant and researcher. Teaching assistant for Practical Python Programming and the Algorithms and Computational Thinking courses, instructing 500+ students. - Maintain the data-driven surveys website, datadrivensurvey.com — published in ACM IMWUT.	Feb 2025 — Present <i>Lausanne, Switzerland</i>
Master Thesis Irmos Technologies AG - Applied unsupervised AI methods for continuous structural monitoring of bridges using vibration time-series data. - Built a physical bridge model in ANSYS WB to determine modal shapes — published in Sensors MDPI.	2024 <i>Zurich, Switzerland</i>
Fullstack Developer ETH Juniors - Developed a photogrammetric application using C#, JavaScript, HTML, Python, Docker, and Azure. - Reduced backend processing time by approximately 23%; designed architecture and identified security risks. - Developed geometric calculation methods and applied statistical techniques to improve 3D models.	Oct 2022 — Jan 2023 <i>Zurich, Switzerland</i>
Student Assistant École Polytechnique Fédérale de Lausanne Teaching assistant in Analysis at EPFL for Prof. Dr. Anna Lachowska.	Sept 2019 — Dec 2019 <i>Lausanne, Switzerland</i>
Summer Internship in Brain Imaging Dawson College & Brain Imaging Centre Worked on phase correction for MRI signal acquisition under Prof. Dr. Nadeau.	May 2015 — Aug 2015 <i>Montréal, Canada</i>

PROJECTS

Ethereum Smart Contract Security Analysis - Security analysis of smart contracts using 1,000 lines of Datalog. Grade: 6.0. - Source: github.com/AmroAbdrabo/ethereum-sc-taint-analyzer.	Feb 2022 — June 2022
VOJ – Voice of Jungle - ETH AI Center Hackathon winner; identified bird species from acoustic data using ML. - Trained a wav2vec2 transformer on Xeno Canto data. Source code: github.com/AmroAbdrabo/gainforest-voj.	May 2024
OpenEye Entrepreneur First Hackathon winner; early disease detection via AI-powered retinal scanners with medical data anonymization.	Apr 2024
Catrine 3D puzzle game built with MonoGame (C#) and a C++ physics engine. Trailer: youtu.be/WIuS6iGwTIA.	Feb 2022 — June 2022
Electricity Consumption Forecasting Datathon project with the third-lowest error forecasting ALPIQ electricity consumption in Spain and Italy.	May 2024

AWARDS

- Peter Berlow Award, Dawson College:** Highest average in first year of Pure and Applied Sciences (among 300+ students).
- AI for Good Hackathon, Entrepreneur First / Visium:** 1st place, organized by Entrepreneur First, Visium and EPFL AI Team.
- Best Project – EcoHackathon, ETH AI Center / Gainforest:** Bird species identification from acoustic data using wav2vec2 on Xeno Canto.

SKILLS

- Programming and tools:** Python (PyTorch), Java, React, SQL, REST, Blender, C#, JavaScript, HTML, Docker, Azure.
- Languages:** English (native/bilingual), German (full professional), Arabic (full professional), French (full professional).